

THRONIX BACKEND — PRODUCT REQUIREMENTS DOCUMENT (PHASE 1)

Schedule A to the Software Development Agreement (Backend)

1. Design principles (govern the whole system)

- **Human-in-the-loop, permanently.** The AI layers analyse, predict, and recommend; a human decides and acts on anything that affects the field or any safety response. The AI never takes autonomous action and is never in the safety-action path. Life-safety decisions are always taken by a person. This is a fixed design boundary, not a phase that is later relaxed.
- **Deterministic safety floor, independent of AI.** The deterministic triggers (H₂S, well/casing integrity, kick precursor, ESD) run independently and cannot be suppressed, delayed, gated, or overridden by the AI layers. Consistent with the THRONIX Spec, H₂S is suppressed by nothing and automatic changes may only lower autonomy, never raise it.
- **AI sits alongside and on top, read-and-recommend only.** The predictive and agentic layers read live data, the deterministic outputs, and the analytics, and produce explainable recommendations. They do not write to control, change setpoints, or alter safety logic.

2. Scope summary

Phase 1 delivers, as the backend:

- the THRONIX deterministic SAFE-CORE system — telemetry ingestion, validation, the well-state machine, the trigger engine (T1/T2/T3), gas response, mode/degradation logic, ring escalation, and alert output, per the THRONIX Spec;
- a leak-proof data-capture backbone — durable, access-controlled storage of incoming signals and of the actions/decisions taken, feeding the analytics layer and retained for later phases;
- a predictive and analytical AI layer over the live and captured data;
- an advisory agentic AI layer that reasons over the above and recommends actions to a human (recommend-only);
- a per-well sensor-health monitoring view;
- automated reporting (scheduled reports auto-sent to configured recipients); and
- the replay/validation harness used to prove the triggers and logic before they arm.

Further specifications requested by stakeholders are added to this Schedule A by written agreement of the Parties.

3. Functional requirements — deterministic backend (SAFE CORE)

Built to the THRONIX Spec. The following must be implemented and verifiable:

- Telemetry contract (Spec §2): ingest the live channels, per-well reference data, and site reference data; honour per-channel `instrument_span` and `slew_max`; any trigger whose reference field is missing declares itself UNARMED + reason and never silently passes.

- Layer-0 validation (§3.1): median (Hampel) despiking; range, slew, frozen, and staleness checks; failures quarantine to proxies; slew suspended around registered state changes; raised temperature span honoured for thermal wells.
- Survivability (§3.2–3.7): H₂S sensor-honesty rules; split gap ladders for safety vs production channels; degradation_score and VERIFIED → DEGRADED → SURVIVAL mode mapping with hysteresis; sensor-health classes; comms-latency bands; edge vs central split (SAFE CORE survives comms loss).
- Well-state machine & state trust (§4): the seven states; UNKNOWN → FLOWING set-points (fail-armed); state-probability fusion and the trust ladder; suppression only on real signature with sufficient probability; a declared state contradicting physics reverts to FLOWING and raises a security alert; H₂S suppressed by nothing.
- Trigger engine (§6): T1 safety-critical (incl. the H₂S area-safety and change-detection arms, integrity, kick precursor, ESP, theft); T2 slow/statistical (flowing-only, advisory until window filled); T3 24-h; cold-start/SEEDED behaviour; anchored AND rolling baselines; windowed-regression rate; per-well and baseline-relative thresholds where the Spec requires.
- Alert architecture (§5, §11): budgets and flood control (EEMUA 191 / ISA-18.2); grouping; tiered gas response with wind/muster guidance; the alert response format with data sources, limit crossed, well state, data-quality context, and recommended action.
- Baselines & re-anchoring (§7); coarse confidence & explainability (§8); the Simplification Governor & SAFE CORE tagging (§12).
- All prohibited behaviours (§14) are enforced; all go-live gates (§17), including the H₂S breathing-zone and thermal-well gates, are respected (a gated well stays NOT-CLEARED / UNARMED rather than arming unsafely).
- Sensor-health monitoring view: a per-well view of sensor health using the Spec's sensor-health classes (§3.6), surfacing degraded, drifting, suspect, failed, stale, or calibration-overdue sensors with their data-quality and calibration-age context — degraded or missing sensors are surfaced, never hidden (§14).
- Automated reporting: scheduled reports (e.g., shift, alert, and sensor-health summaries) generated and auto-sent to configured recipients on a defined schedule.

4. Functional requirements — data-capture backbone

The system captures and retains, durably and queryably, the data needed by the analytics layer and by later phases:

- Signals: ingested telemetry and per-sample data-quality/state context (sensor_health, gap, degradation_score, well_state, baseline deltas).
- Actions & decisions: every alert and its full lifecycle, plus the operator actions/overrides taken (Spec §9) — so the record captures what the signals were and what action was taken.
- Provenance: the per-sample signed hash where signing exists; otherwise the provenance flag, recorded with each record.
- The capture is structured and labelled for analysis and for later model training; bespoke model training on this data is a later phase (Section 8).

5. Data security — leak-proof requirement

- Encryption in transit (TLS) and at rest for all stored signals, alerts, and action logs, including backups.
- Access control: least-privilege, role-based access; no shared credentials; secrets held in a secrets manager / environment, never in source code or the frontend.
- Auditability: access to the data store is logged; stored records are tamper-evident (append-only or hash-chained where practicable).
- No third-party leakage: data is not sent to, retained by, or used to train any external AI/ML or third-party service except as expressly permitted under the NDA's Approved AI / Development Tools schedule on a non-retention / non-training configuration.
- Compliance with the data-protection and breach-notification duties in the Parties' NDA / IP-Assignment Agreement.

6. Predictive & analytical AI layer (Phase 1)

An analytical layer over the live and captured data that produces insight, not action:

- Analytics across wells and time — trends, anomalies, and condition summaries built on the deterministic outputs and the captured history.
- Predictive signals — early indication of developing conditions (e.g., trend-based deterioration), expressed with explainable rationale and a coarse, honestly-labelled confidence (Spec §8); no claim of calibrated probability where none exists.
- Prioritisation — surfacing what matters now from the deterministic alert stream, without throttling detection or logging.
- **Maturity note.** the layer is built on the data available at go-live; deeper model-based capabilities (e.g., physics/ML twins, remaining-useful-life, causal reasoning) mature as operational-data history accrues, exactly as the THRONIX Spec Section 16 preconditions describe. Building those bespoke models on accumulated proprietary data is a later phase (Section 8).

This layer never acts: it informs the deterministic system's human operators and the agentic layer's recommendations.

7. Advisory agentic AI layer (Phase 1)

An agent that reasons over the deterministic outputs, the analytics, and the predictions, and produces recommended actions for a human — working alongside, not replacing, the hardcoded thresholds:

- Triage & investigation — group related alerts, gather corroborating evidence (sister-well readings, recent baselines, state context), and present a tied-up case rather than raw alerts.
- Diagnosis & recommendation — explain what is happening and recommend an action, with the supporting signals and the rationale.
- **Strictly advisory.** the agent recommends; a human decides and executes. It takes no autonomous action, issues no control or setpoint change, and is never in the safety-action path. It cannot suppress, delay, or gate any deterministic safety trigger.
- **Explainable & validated.** every recommendation states its basis (driving signals, limits, deltas) so an operator can judge it; the agent's outputs are exercised in the validation harness (Section 9) so a poor recommendation cannot quietly mislead.

8. Non-functional requirements

- Scale/cadence: handle the Spec's target of up to [~4,000] wells at a 3s sample rate [confirm the Phase 1 target scale, which may be a defined subset for the pilot].
- Edge survivability: SAFE CORE continues to function during comms loss (Spec §6.9, §12); the AI layers are non-blocking — if they are unavailable, deterministic detection and alerting continue unaffected.
- Reliability: timing rules use wall-clock plus a minimum count of valid samples; fast safety channels at sample rate, slow channels on 1-minute averages.
- Tech stack / hosting / repository: [confirm — language(s), datastore, deployment target, shared repo].
- Source code, configuration, and documentation delivered to the Client's repository; complete dependency list with licences at handover (Schedule C of the Agreement).

9. Boundaries — excluded and later

Permanently excluded by design (not a future phase):

- Any autonomous or closed-loop action by the AI; any AI control of equipment, setpoints, chokes, or shut-in.
- Any AI suppression, triggering, or gating of a safety response (H₂S muster, ESD, integrity); any AI decision over life-safety. These remain human, over the deterministic floor.

Later / separate (commissioned as a separate engagement or change order under Clause 9.2):

- Training of bespoke models on accumulated proprietary operational data (e.g., physics/ML twin, remaining-useful-life, causal-graph reasoning) — gated by the data history the THRONIX Spec Section 16 requires.
- Other automations beyond Phase 1 reporting (e.g., auto-drone dispatch, natural-language operator overrides).
- Third-party certification.

10. Acceptance criteria (objective)

Build Acceptance — the build is accepted when:

- the deterministic backend (Section 3) is implemented per the THRONIX (previously known as oasis) Spec and runs end-to-end on a fresh deployment without error;
- the replay / validation harness passes: per-trigger and per-well-state precision, recall, and lead-time are reported, plus the adversarial replays (spoofed state, frozen value, replay, theft); no trigger arms live ESD before passing replay; missing inputs surface as UNARMED + reason;
- the data-capture backbone (Section 4) records signals and actions and the records are retrievable;
- take the proxy system into consideration as well when the sensor(s) are down
- the leak-proof security requirements (Section 5) are met — encryption in transit and at rest, role-based access, no secrets in code or frontend, access logging, and no external-tool leakage;

- the predictive/analytical layer (Section 6) produces its outputs with explainable rationale and labelled coarse confidence; and
- the agentic layer (Section 7) produces explainable recommendations and is verified to be advisory only — it cannot suppress, delay, gate, or override any deterministic safety trigger, and issues no autonomous action;
- the per-well sensor-health view reflects the Spec's sensor-health classes and surfaces degraded, failed, stale, or calibration-overdue sensors; and
- scheduled reports are generated and auto-delivered to the configured recipients.
- all the specification in the framework which is the core of all operations

Final Acceptance — occurs on successful completion of the pilot phase with the above stable in operation (or the Longstop Date in Schedule B), per Clause 4 of the Agreement.

11. Assumptions & dependencies

- The Client supplies the per-well, per-channel, per-detector, and per-site reference data and go-live inputs listed in THRONIX Spec Section 17; a trigger stays UNARMED + reason until its inputs are supplied.
- The Client provides telemetry access / feeds (or a representative dataset for the pilot) and the labelled datasets needed to run the replay harness.
- Real cryptographic signing / provenance crypto is a Spec Section 16 item; until deployed, the system uses the provenance flag and no state earns full suppression (Spec §4.3).
- The Client provides feedback within the review windows in Clause 4 / Schedule B of the Agreement.

12. Sign-off

The Parties confirm this PRD is the agreed Schedule A scope and acceptance criteria for Phase 1.

Client (Thronix): POOJA VINAY SURYAVANSHI Date: _____

Developer: _____ Date: _____